Software Engineering

LAB 4: UML SEQUENCE / STATE /ACTIVITY DIAGRAMS

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Learning Objectives

- Students will learn to translate a narrative process into a formal UML Sequence State Activity Diagram.
- They will identify Actors, Lifelines, and Messages to visualise how objects interact over time.

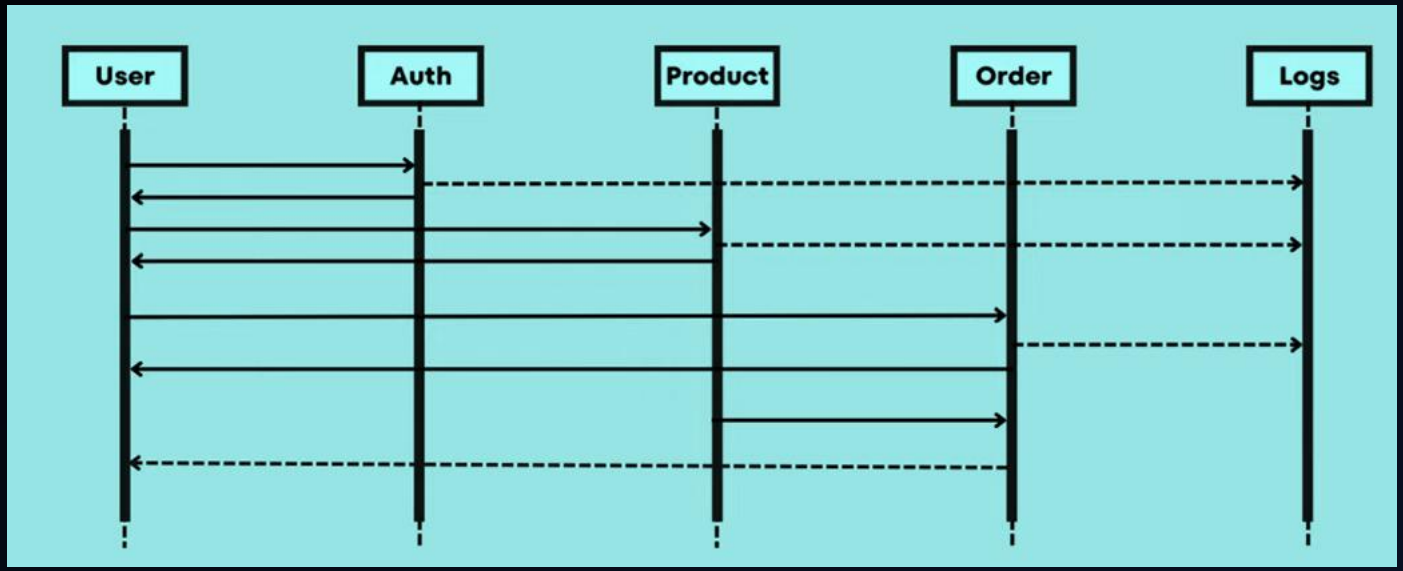
Core Components to Identify

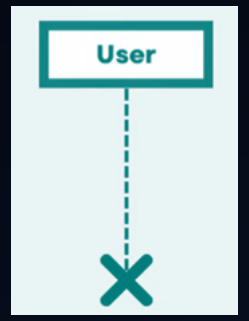
- Actors: External entities (usually a person) that initiate the process.
- Lifelines: Vertical dashed lines representing the existence of an object/system over time.
- Activation Bars: Thin rectangles on lifelines showing when a process is "active.“
- Synchronous Messages: Solid lines with solid arrows (waits for a response).
- Return Messages: Dashed lines with open arrows (returning data/control).

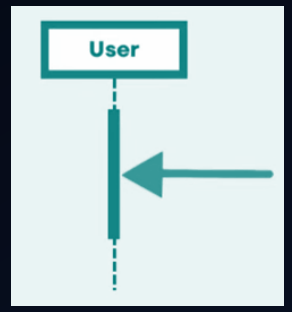
Core Components to Identify

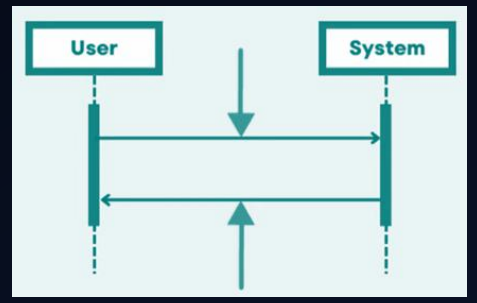
- **Synchronous Message:** Solid line with a **filled** arrowhead (sender waits for response)
- **Asynchronous Message:** Solid line with an **open** arrowhead (sender does not wait, e.g., sending an email)
- **Self-Message:** A loop-back arrow used for internal system validation
- **Create/Destroy:** Using a "Create" arrow to generate a new object and an "X" to terminate a lifeline (e.g., ending a session)
- **Found/Lost Messages:** Identifying interactions that enter or leave the system from unknown origins

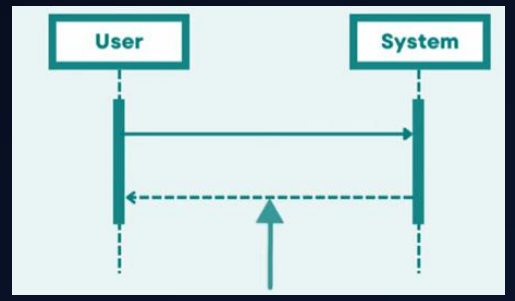
Core Components to Identify

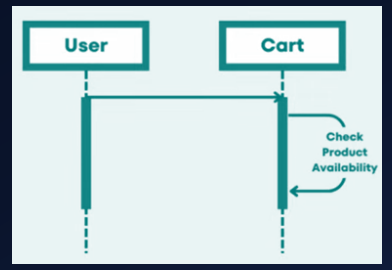








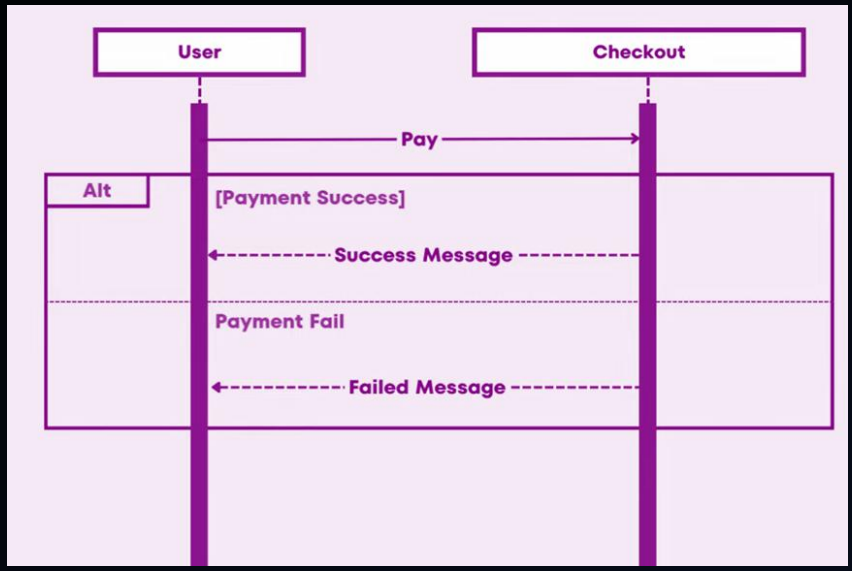


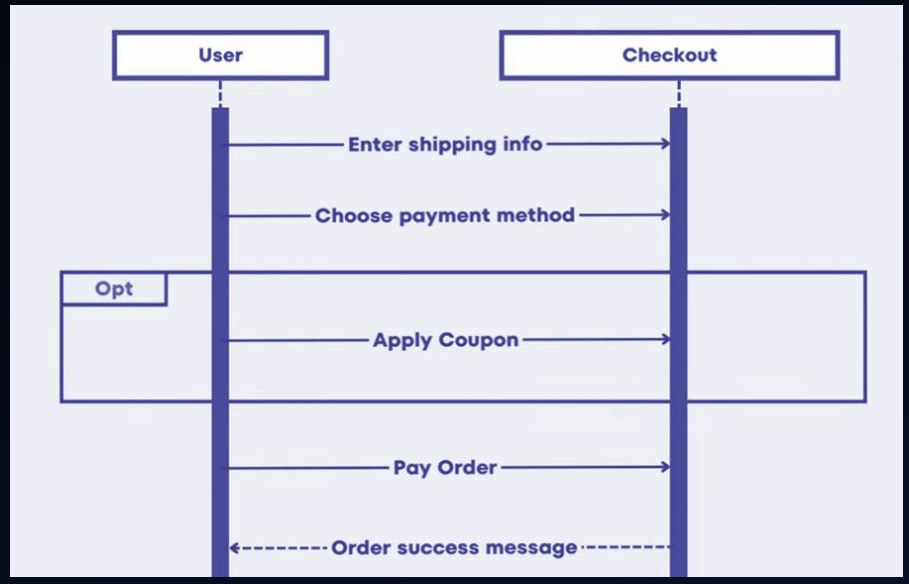


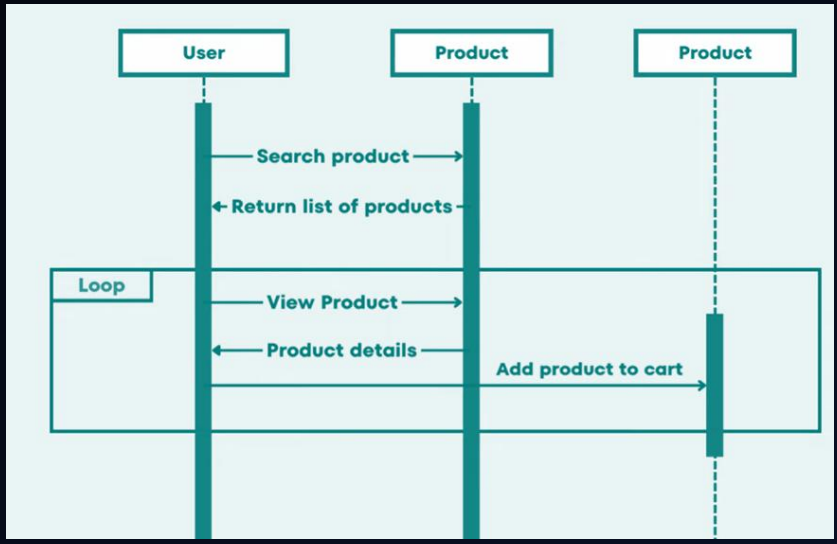
Advanced Logic (Combined Fragments)

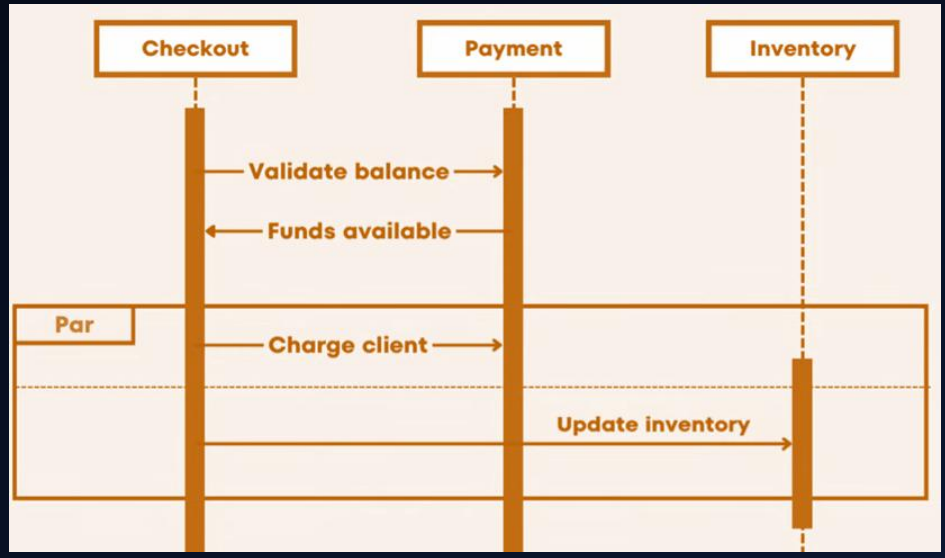
The lab report must explicitly label and explain the use of these "Combined Fragments“

- alt: For conditional branching (e.g., Payment Success vs. Failure
- opt: For steps that might not happen (e.g., Applying a discount code).
- loop: For repetitive actions (e.g., adding multiple items to a cart).
- par: For simultaneous actions (e.g., updating inventory while processing payment)


User
Checkout
Pay
Alt
[Payment Success]
Success Message
Payment Fail
Failed Message


User
Checkout
Enter shipping info
Choose payment method
Opt
Apply Coupon
Pay Order
Order success message


User
Product
Product
Search product
Return list of products
Loop
View Product
Product details
Add product to cart


Checkout
Payment
Inventory
Validate balance
Funds available
Par
Charge client
Update inventory

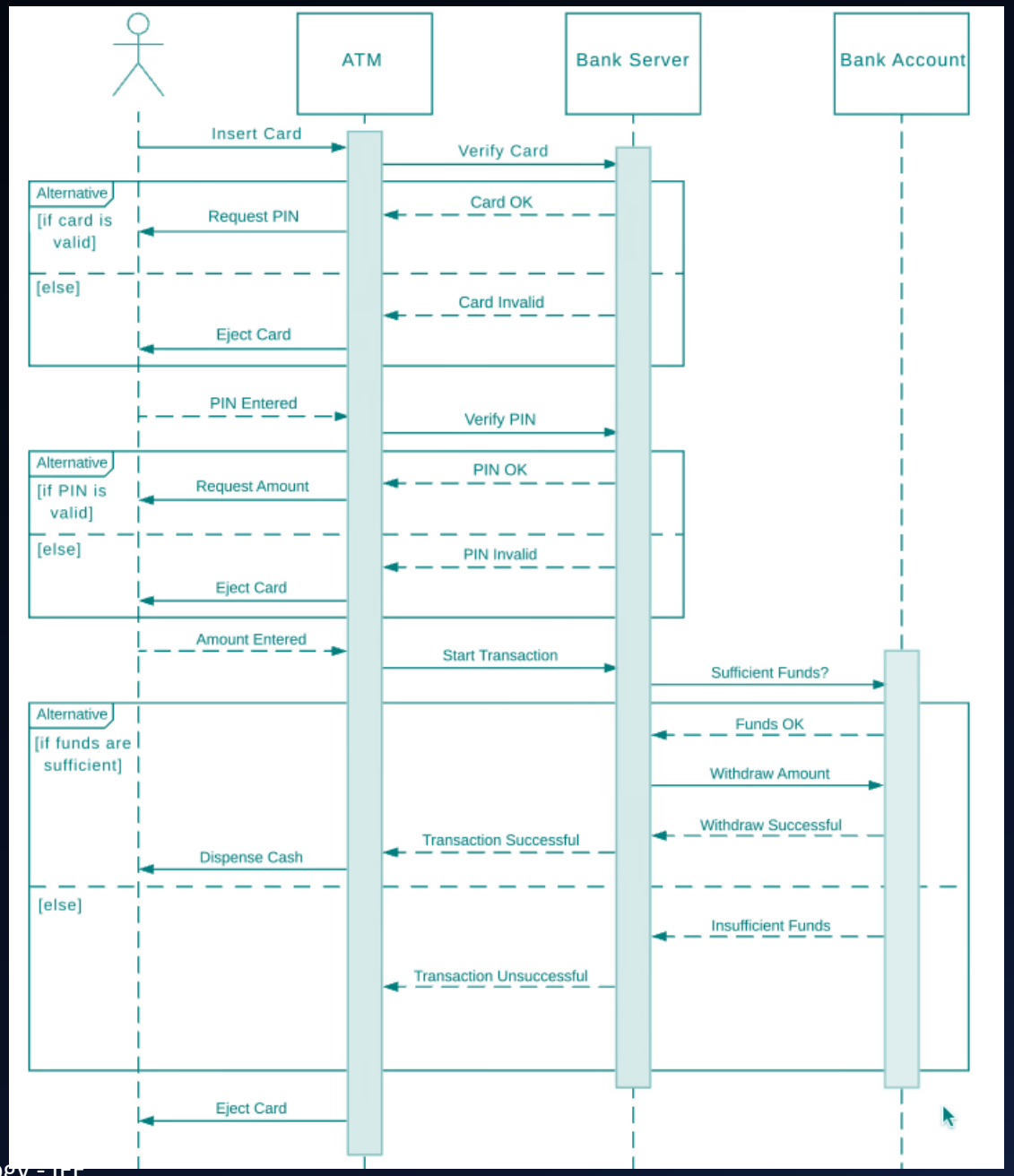
The Core Challenge – Online Shopping Experience

Participants (Lifelines) to Include:

- **User/Customer**
- **Login Module**
- **Product Catalog**
- **Shopping Cart**
- **Checkout & Payment** (External/Third-Party)

Step-by-Step Instructions

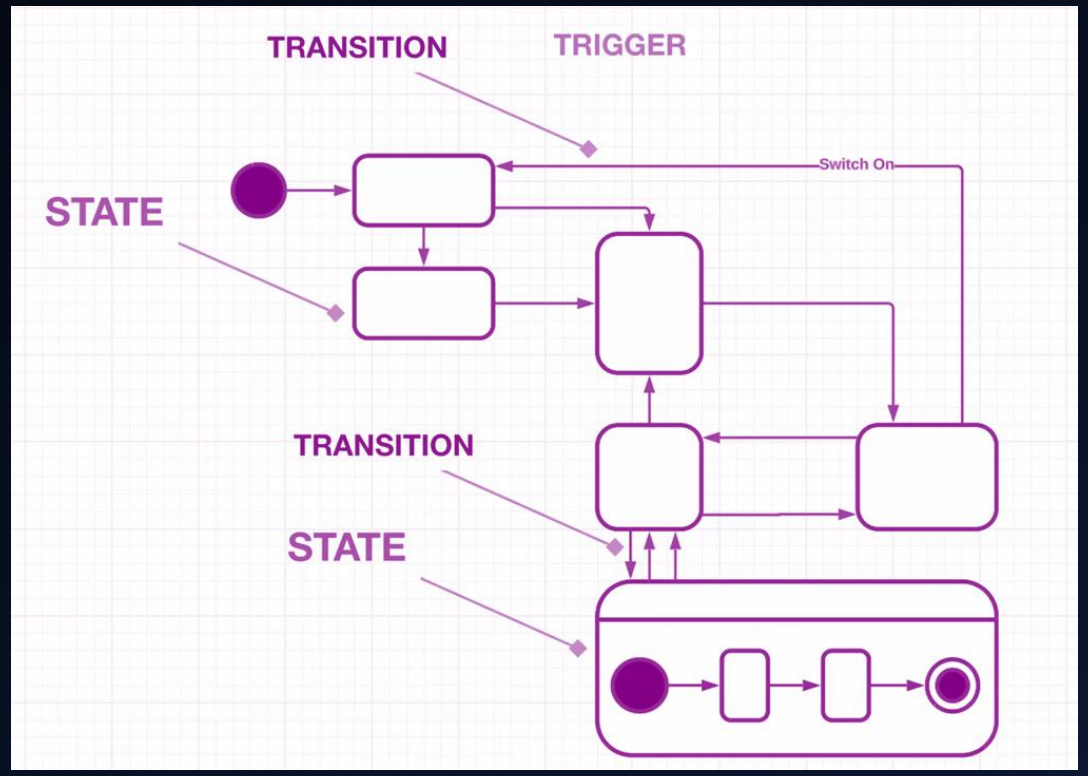
Step	Action	Description
1	**Identify Participants**	List all Actors (e.g., Customer) and Systems (e.g., ATM, Bank Database).
2	**Set the Timeline**	Draw lifelines for each participant from left to right.
3	**Map Messages**	Draw horizontal arrows to represent requests and dashed arrows for responses.
4	**Add Logic (Alt/Loop)**	Use a "Combined Fragment" (an alt box) to show what happens if a PIN is incorrect.
5	**Peer Review**	Have students check if the arrows follow a logical top-to-bottom time sequence.



State Diagrams

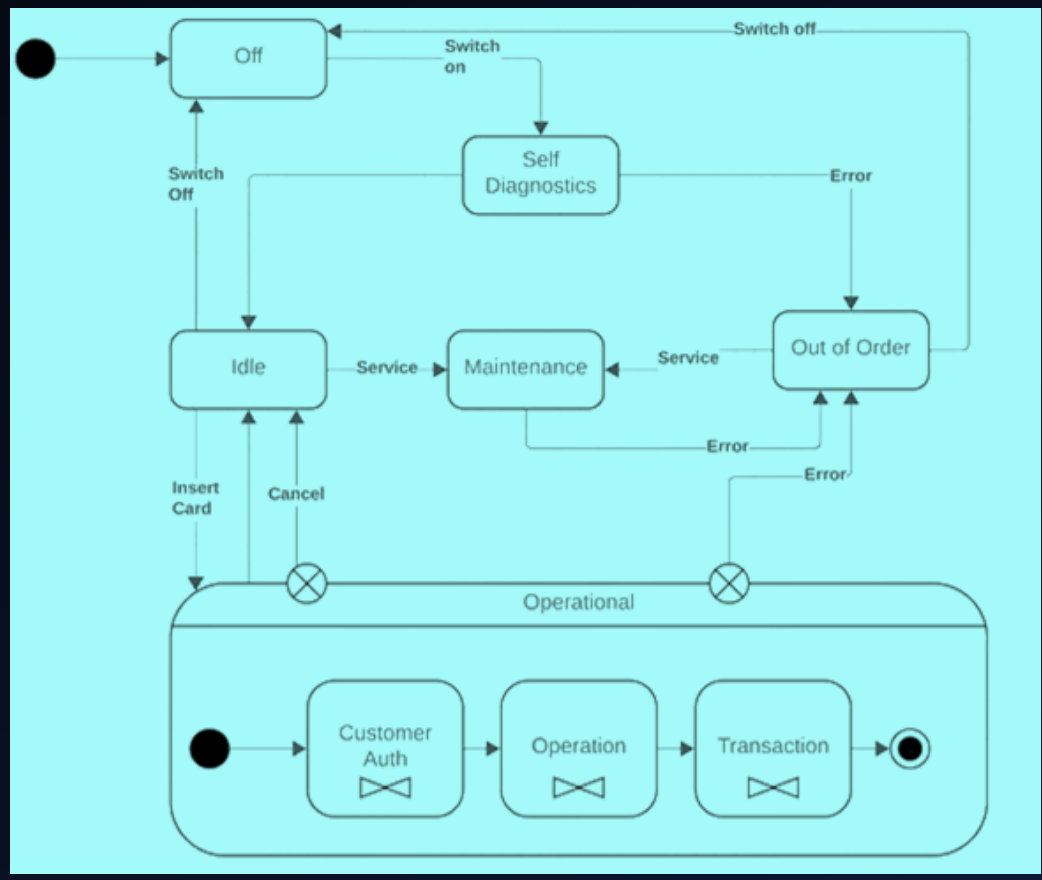
Key Concepts & Notation

- State: A condition or situation during the life of an object (e.g., Idle, Out of Service).
- Transition: A relationship between two states indicating that an object will perform certain actions and move to the second state.
- Initial Pseudo-State: The starting point of the diagram (solid black circle).Final Pseudo-State: The end of the process (a bullseye symbol).
- Composite State: A state that contains substates (e.g., "Operational" containing "Authentication").



Identifying Top-Level States

- **Off:** The starting point before the machine is powered on.
- **Self-Diagnostics:** Triggered immediately after being switched on.
- **Idle:** Waiting for a customer to insert a card.
- **Operational:** The complex state where the user interacts with the machine.
- **Out of Service:** When a failure occurs.
- **Maintenance:** When a technician is fixing or updating the system.



Modeling Transitions & Triggers

Draw arrows to connect the states and label them with the triggering events:

•**Power On** → Leads from *Off* to *Self-Diagnostics*.

•**Success** vs. **Failure** → Leads from *Diagnostics* to *Idle* OR *Out of Service*.

•**Insert Card** → Leads from *Idle* to *Operational*.

•**Cancel/Finish** → Leads from *Operational* back to *Idle*.

•**System Error** → Leads from *Operational* to *Out of Service*.

Activity Diagrams

Core Components to Use

- Initial Node: A solid black circle where the flow begins
- Action Nodes: Rounded rectangles representing a single step in the process (e.g., "Create Order").
- Control Flows (Edges): Arrows showing the sequence of actions
- Decision & Merge Nodes: Diamond shapes used to show branching paths (e.g., Credit Card vs Cash).
- Fork & Join: Thick horizontal or vertical bars used to show parallel actions that must happen simultaneously.
- Final Nodes: The "Bullseye" (Activity Final) to end the entire process, or the "X" (Flow Final) to end one specific path.

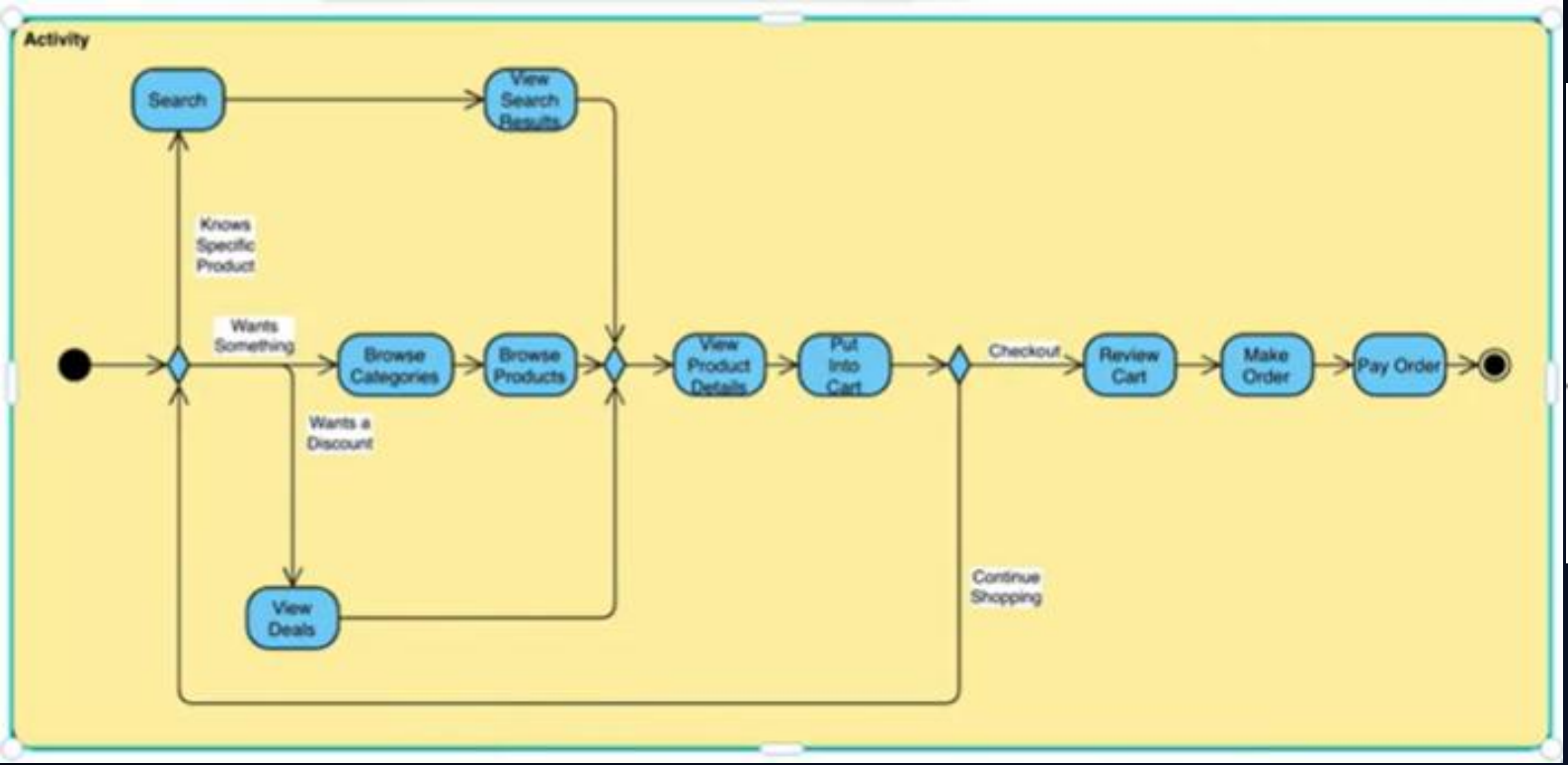
Online Shopping

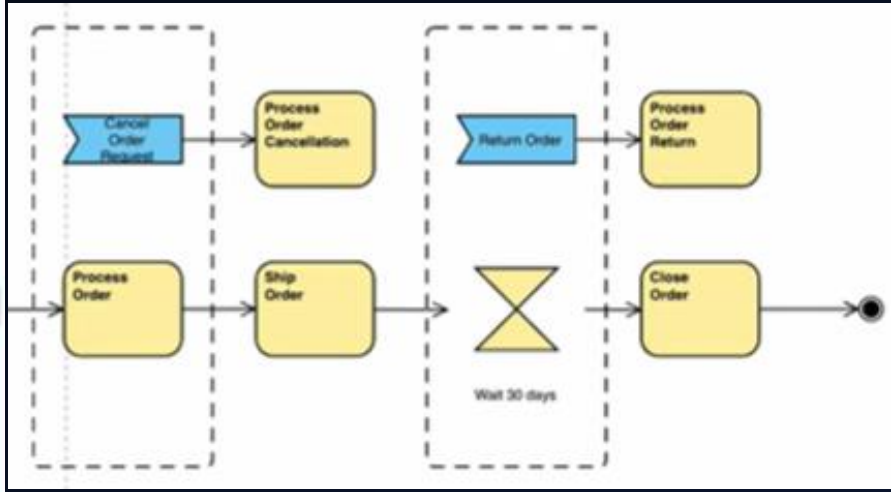
- Start: The user searches for a product.
- Selection: The user browses categories or specific search results
- Cart Action: The user selects a product and adds it to the cart
- Checkout Logic (Decision): The system asks for the payment type.
  - If Credit Card: Process the payment and generate an invoice.
  - If Cash: Move directly to order confirmation.
  - Parallel Processing (Fork/Join): While the order is being packed, the system must simultaneously send a confirmation email
  - End: The process completes once the order is shipped.

Step-by-Step

Step	Action	Description
1	**Define the Scope**	Use a large rounded rectangle with the name "Buy Product" in the top-left corner.
2	**Map the Main Path**	Use Action Nodes for "Browse," "Add to Cart," and "Checkout."
3	**Add Decision Guards**	Use a diamond for payment selection. Label the arrows (Guards) with "Cash" and "Credit Card".
4	**Model Concurrency**	Use a **Fork** bar to show the "Pack Order" and "Notify Customer" actions happening at the same time.
5	**Advanced Feature**	Add an **Interruptible Region** (dashed box) to show where a user can "Cancel Order" at any time before shipment.

Step-by-Step


Activity
Search
View Search Results
Knows Specific Product
Wants Something
Browse Categories
Browse Products
View Product Details
Put Into Cart
Checkout
Review Cart
Make Order
Pay Order
Wants a Discount
View Deals
Continue Shopping


Cancel Order Request
Process Order Cancellation
Return Order
Process Order Return
Process Order
Ship Order
Wait 30 days
Close Order

Task

Implement the following systems

System 1: ATM (Automated Teller Machine) : Sequence diagram

System 2: Medical Clinic System: State diagram

System 3: Car Insurance System: Activity

Remember to update your SRS Documents to include your topic with diagrams.